SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	VAIBHAVA LAKSHMI K	**Reg. No.:**	192511310
Section / Batch:	2025-2029	**Date:**	22/07/2026

Code of Conduct

I VAIBHAVA LAKSHMI K (192511310) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

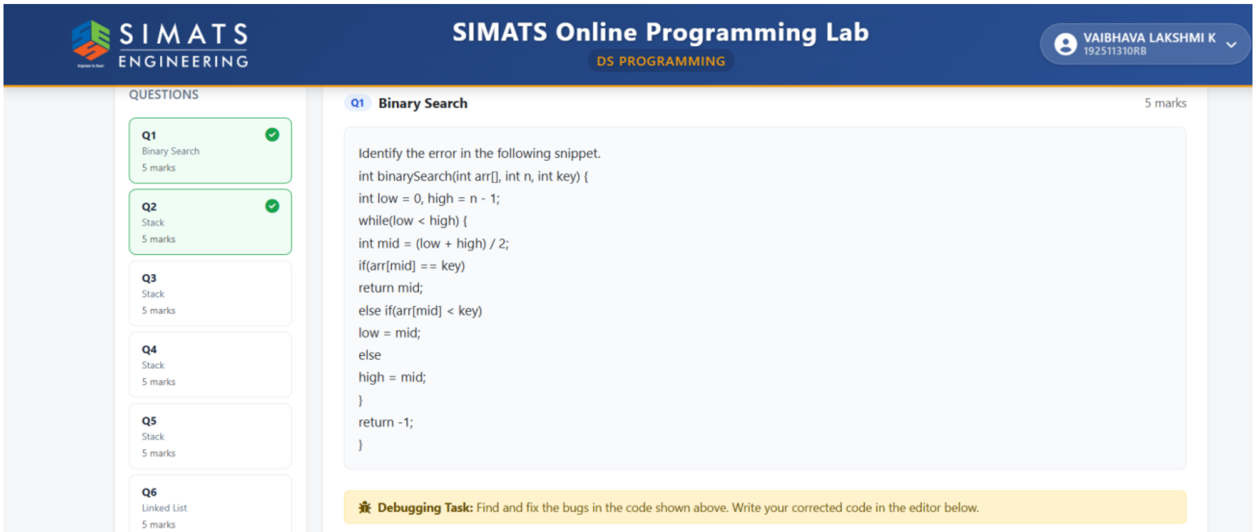
Signature of the Student: VAIBHAVA LAKSHMI K

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
		GRAND TOTAL:	**100 Marks**

DEBUGGING: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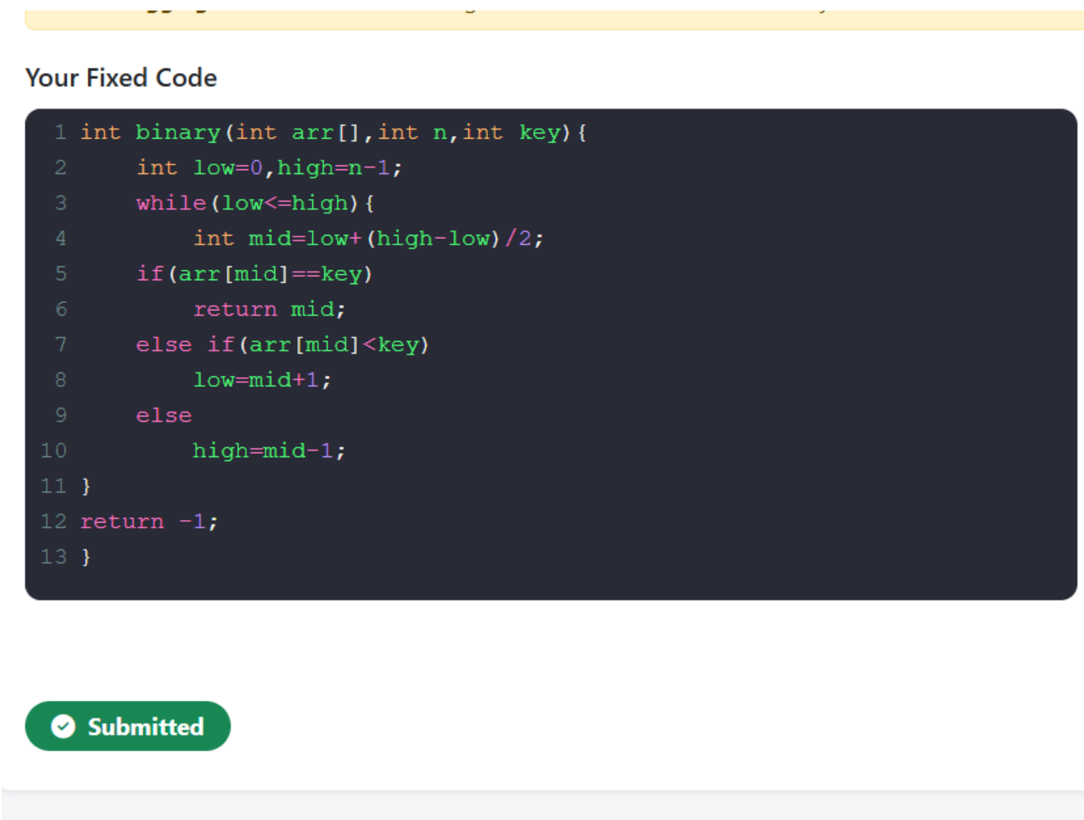
QUESTIONS
Q1
Binary Search
5 marks
Q2
Stack
5 marks
Q3
Stack
5 marks
Q4
Stack
5 marks
Q5
Stack
5 marks
Q6
Linked List
5 marks
Q1 Binary Search
5 marks
Identify the error in the following snippet.
int binarySearch(int arr[], int n, int key) {
int low = 0, high = n - 1;
while(low < high) {
int mid = (low + high) / 2;
if(arr[mid] == key)
return mid;
else if(arr[mid] < key)
low = mid;
else
high = mid;
}
return -1;
}
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.


Your Fixed Code
int binary(int arr[],int n,int key){
int low=0,high=n-1;
while(low<=high){
int mid=low+(high-low)/2;
if(arr[mid]==key)
return mid;
else if(arr[mid]<key)
low=mid+1;
else
high=mid-1;
}
return -1;
}
Submitted

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack

Q2 Stack 5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int pop(){
    if (top==-1){
        printf("Underflow\n");
        return -1;
    }
    return stack[top--]
}
```

Test Input

stdin input (optional)

Output

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks

Q3 Stack 5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int stack[5];
int top;

void init() {
    top = -1;
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks

Q4 Stack 5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
while(stack[top] != ')') {
postfix[j++] = pop();
}
pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if(ch == ')') {
    while(stack[top] != '(') {
        postfix[j++] = pop();
    }
    pop();
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks

Q5 **Stack** 5 marks

In balancing Bracket, you identify the error.
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
pop();
} else {
return 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if (top != -1 && ((ch == ']' && stack[top] == '[') ||
                  (ch == '}' && stack[top] == '{') ||
                  (ch == ')' && stack[top] == '('))) {
    pop();
} else {
    return 0;
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List

Q6 **Linked List** 5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
temp = head
while temp != None:
    print(temp.data)
    temp = temp.next
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks

Q7 Stack 5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
POP(stack) {
    if (top == -1) {
        printf("Underflow\n");
        return -1; // Or error handling
    } else {
        return stack[top--];
    }
}
```

Test Input

stdin input (optional)

Output

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7

Q8 Linked List Data Structures 5 marks

Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
prev = None
current = head

while current is not None:
    next_node = current.next
    current.next = prev
    prev = current
    current = next_node

head = prev
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack

Q9 Linked List 5 marks

What is the error in the below code?
new.next = head
head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
new_node->next = head;
head = new_node;
```

Test Input

stdin input (optional)

Output

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks

Q10 **Queue** 5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int DEQUEUE(int Q[]) {
    if (front == -1 || front > rear) {
        return -1;
    }
    int item = Q[front];
    if (front == rear) {
        front = -1;
        rear = -1;
    } else {
        front++;
    }
    return item;
}
```

Test Input

stdin input (optional)

Output

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- Stack Notations - Postfix Expression 5 marks
- Q2 Circular Queue 5 marks
- Q3 Binary Search 5 marks
- Q4 Queue 5 marks
- Q5 Queue 5 marks
- Q6 Doubly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Stack 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack 5 marks

Find the bug in the evaluation of a Postfix expression.
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
while (expression[i] != '\0') {
    if (isdigit(expression[i])) {
        push(expression[i] - '0');
    } else {
        int op2 = pop();
        int op1 = pop();
        int result;

        switch (expression[i]) {
            case '+': result = op1 + op2; break;
            case '-': result = op1 - op2; break;
            case '*': result = op1 * op2; break;
            case '/': result = op1 / op2; break;
        }
        push(result);
    }
    i++;
}
```

Test Input

stdin input (optional)

Output

Submitted

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QUESTIONS

- **Q1** Stack Notations - Postfix Expression — 5 marks
- **Q2** Circular Queue — 5 marks
- **Q3** Binary Search — 5 marks
- **Q4** Queue — 5 marks
- **Q5** Queue — 5 marks
- **Q6** Doubly Linked List — 5 marks
- **Q7** Stack — 5 marks

Q2 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int dequeue() {
    if (front == -1) return -1;
    int data = queue[front];
    if (front == rear) {
        front = -1;
        rear = -1;
    } else {
        front = (front + 1) % MAX;
    }
    return data;
}
```

Test Input

stdin input (optional)

Output

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QUESTIONS

- **Q1** Stack Notations - Postfix Expression — 5 marks
- **Q2** Circular Queue — 5 marks
- **Q3** Binary Search — 5 marks
- **Q4** Queue — 5 marks
- **Q5** Queue — 5 marks
- **Q6** Doubly Linked List — 5 marks
- **Q7** Stack

Q3 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.
int binarySearch(int arr[], int l, int r, int x) {
int mid = l + (r - l) / 2;
if (arr[mid] == x) return mid;
if (arr[mid] > x) return binarySearch(arr, l, mid, x);
return binarySearch(arr, mid, r, x);
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int binarySearch(int arr[], int l, int r, int x) {
    if (r >= l) {
        int mid = l + (r - l) / 2;
        if (arr[mid] == x) return mid;
        if (arr[mid] > x) return binarySearch(arr, l, mid - 1, x);
        return binarySearch(arr, mid + 1, r, x);
    }
    return -1;
}
```

Test Input

stdin input (optional)

Output

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- **Q1** Stack Notations - Postfix Expression — 5 marks
- **Q2** Circular Queue — 5 marks
- **Q3** Binary Search — 5 marks
- **Q4** Queue — 5 marks
- **Q5** Queue — 5 marks
- **Q6** Doubly Linked List — 5 marks
- **Q7** Stack — 5 marks

Q4 Queue

5 marks

Find the error in this Deque insertFront operation.
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void insertFront(int data) {
    if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {
        return;
    }
    if (front == -1) {
        front = 0;
        rear = 0;
    } else if (front == 0) {
        front = MAX - 1;
    } else {
        front = front - 1;
    }
    deque[front] = data;
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- **Q1** Stack Notations - Postfix Expression — 5 marks
- **Q2** Circular Queue — 5 marks
- **Q3** Binary Search — 5 marks
- **Q4** Queue — 5 marks
- **Q5** Queue — 5 marks

Q5 **Queue** 5 marks

Identify the error in the Queue enqueue logic.
void enqueue(int data) {
if (rear == MAX) return;
queue[rear++] = data;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void enqueue(int data) {
    if (rear == MAX - 1) return;
    if (front == -1) front = 0;
    queue[++rear] = data;
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

- **Q1** Stack Notations - Postfix Expression — 5 marks
- **Q2** Circular Queue — 5 marks
- **Q3** Binary Search — 5 marks
- **Q4** Queue — 5 marks
- **Q5** Queue — 5 marks
- **Q6** Doubly Linked List — 5 marks

Q6 **Doubly Linked List** 5 marks

Identify the error in this Doubly Linked List node deletion.
void deleteNode(Node* del) {
if (del->prev != NULL)
del->prev->next = del->next;
if (del->next != NULL)
del->next->prev = del->prev;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void deleteNode(Node* del) {
    if (del == NULL) return;
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
    free(del);
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- **Q1** Stack Notations - Postfix Expression — 5 marks
- **Q2** Circular Queue — 5 marks
- **Q3** Binary Search — 5 marks
- **Q4** Queue — 5 marks
- **Q5** Queue — 5 marks

Q7 **Stack** 5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int pop() {
    if (top == -1) {
        return -1;
    }
    return stack[top--];
}
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack Notations - Postfix Expression 5 marks
- Q2 Circular Queue 5 marks
- Q3 Binary Search 5 marks
- Q4 Queue 5 marks
- Q5 Queue

Q8 **Stack** 5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void push(int data) {
    if (top == MAX - 1) {
        return;
    }
    stack[++top] = data;
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack Notations - Postfix Expression 5 marks
- Q2 Circular Queue 5 marks
- Q3 Binary Search 5 marks
- Q4 Queue 5 marks
- Q5 Queue 5 marks
- Q6 Doubly Linked List

Q9 **Circular Queue** 5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if ((rear + 1) % MAX == front) {
    printf("Queue Full");
} else {
    if (front == -1) front = 0;
    rear = (rear + 1) % MAX;
    queue[rear] = vehicle;
}
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

- Q1 Stack Notations - Postfix Expression 5 marks
- Q2 Circular Queue 5 marks
- Q3 Binary Search 5 marks
- Q4 Queue 5 marks
- Q5 Queue 5 marks
- Q6

Q10 **Stack** 5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int peek() {
    if (top == -1) {
        return -1;
    }
    return stack[top];
}
```

Test Input

stdin input (optional)

Output

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________